Can You Beat Randomness?

Football Betting Markets: Walk-Forward Models, Statistical Power, Closing-Line Value and Bankroll Risk

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https://github.com/meh6di/can-you-beat-randomness

October 12, 2026

Abstract

We ask whether statistical models can extract a profit from the pricing of the five major European football leagues, and, more importantly, how much evidence a realistic sample can provide either way. Models (tuned logistic regression, a market–strength log-linear blend, raw and calibrated XGBoost) are retrained every month in a walk-forward design on Elo and exponentially weighted form features that carry over seasons. The betting rule and threshold are chosen on one development season by closing-line value (CLV) against Pinnacle and reported once on an untouched holdout season. Every ROI is accompanied by a bootstrap interval and by a p-value against an efficient-market null in which outcomes are redrawn from de-vigged market probabilities; a power analysis states the smallest edge the sample could detect. Probabilistic quality is assessed with log loss, Brier score and the ranked probability score, and calibration with ECE against its sampling-noise floor. A separate study of the market itself examines favourite–longshot bias, and bankroll risk is quantified by simulation under several staking schemes. No model or betting strategy demonstrates statistically significant outperformance of the market: all holdout ROIs are consistent with the efficient-market null, the optimised blend weight collapses to $w \approx 0$, and the smallest edge the holdout could detect is 22–33% ROI for the 6% threshold strategies (139–226 bets on the full holdout). The logistic regression shows a small, statistically significant pick skill (+2.2%), but its closing-line value remains negative and its holdout ROI is indistinguishable from luck. For educational purposes only; not betting advice.

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1 Introduction

Betting advertising is ubiquitous around football. The question is whether algorithmic models can exploit inefficiencies in the odds, or whether the bookmaker's margin guarantees long-run losses. A tempting but weak way to answer it is to report one season's ROI. With a per-bet standard deviation of roughly 1.3–1.6 units, a sample of a few hundred bets has a standard error of roughly 9–13 percentage points of ROI (Section 6.2), far larger than any plausible edge. This report therefore centres on *inference*: intervals, nulls, power, and a low-variance edge measure (CLV), and only then on profit.

Disclaimer. Solely for educational, academic and research purposes. Not financial or gambling advice.

2 Data

Data come from `football-data.co.uk` (`fetch_data.py`): Premier League, La Liga, Serie A, Bundesliga and Ligue 1, seasons 2019–20 to 2025–26. The 2020–21 season (mostly played without crowds) is used only to build ratings and form, never to train or test. `clean_data.py` validates downloads and rows (decimal odds must exceed 1, duplicates are dropped) and keeps results, goals, shots on target, corners, average, best and Pinnacle odds, and Pinnacle *closing* odds. Closing odds are missing for part of the holdout; this matters for which matches each analysis uses (see the start of Section 6).

3 Features and Leakage Control

Features are built per league across seasons (`engineer_features.py`); the value at a match uses only strictly earlier days.

- **De-vigged market probabilities.** With overround $\Omega = \sum_k 1/o_k$, proportional de-vigging gives $p_k = (1/o_k)/\Omega$. We use the *power* method, $p_k = (1/o_k)^c$ with $c \geq 1$ such that $\sum_k p_k = 1$, which removes proportionally more margin from longshots. The model sees the log-ratios $\log(p_H/p_A)$ and $\log\left(p_D/(p_H + p_A)\right)$, which avoids collinearity.
- **Elo.** Home advantage, a goal-difference multiplier, regression towards the mean between seasons, and lower starting ratings for promoted teams.
- **Exponentially weighted form for both teams:** points, goals for (attack) and against (defence), shots on target for/against, corners for/against, with half-life of five matches, shrunk to the league mean while a team has few games ($n/(n+3)$), plus games-played and rest days.

A unit test perturbs the result of one match and verifies that no feature at or before that match changes.

4 Methodology

4.1 Walk-forward protocol

For every calendar month M of the 2024–25 and 2025–26 seasons: training data are matches before $M - 365$ days; the validation window is the 365 days before M and tunes regularisation, the blend weight and the calibrator; month M is the test set. Models are therefore refit as results arrive and no information after M is used. 2024–25 is the *development* season; 2025–26 is the *holdout*.

4.2 Models

- **Logistic regression:** standardised features, league dummies, C tuned on validation.
- **Blend:** $p \propto p_{\text{mkt}}^{1-w}\, p_{\text{strength}}^{w}$, where p_{strength} comes from a model without market inputs and $w \in [0,1]$ is fitted on validation. $w \approx 0$ means the model adds nothing beyond the market.
- **XGBoost**, raw and recalibrated on the later validation window with a multinomial vector-scaling map.

4.3 Betting rules

One unit is staked per match on the best outcome if its score clears a threshold: *edge* ($\hat{p} - p_{\text{mkt}}$) or *expected value* ($\hat{p} \cdot o - 1$), at average, best-available or Pinnacle prices. Rule, threshold and price source are chosen on the development season (maximum mean CLV, at least 150 bets) and evaluated once on the holdout. The reported strategies use thresholds of 3% and 6%.

5 Statistical Framework

Bootstrap. Bets are resampled with replacement to give percentile intervals for ROI.

Efficient-market null. Holding the placed bets fixed, outcomes are redrawn 20,000 times from the de-vigged market probabilities. The resulting ROI distribution is centred near minus the margin, not zero; the p-value is the share of simulations at least as good as the observed ROI.

Power. With per-bet standard deviation σ and n bets, the smallest true ROI detected with 80% power at 5% (one-sided) is $(z_{0.95} + z_{0.80})\,\sigma/\sqrt{n}$, and a target edge e needs $n = \big((z_{0.95} + z_{0.80})\sigma/e\big)^2$ bets.

Closing-line value. $\text{CLV} = o_{\text{taken}}\, p^{\text{Pinnacle close}} - 1$. It does not depend on who won, so its standard error is an order of magnitude smaller than ROI's. "Pick skill" subtracts the average CLV of all three outcomes of the same match, isolating whether the model chose the better side from the margin every blind bet pays.

Probabilistic scores. Log loss, multiclass Brier score and the ranked probability score (outcomes ordered H<D<A), with paired per-match differences against the market, and expected calibration error (equal-mass bins) compared with the ECE a perfectly calibrated forecaster would show by chance.

6 Results

Two evaluation samples. Analyses that need Pinnacle closing odds (closing-line value, pick skill, and the development/holdout strategy selection) drop every match without them (`clv_analysis.py` and `strategy_select.py`), which leaves 898 holdout matches. All other holdout analyses (ROI and efficient-market null, per-league ROI, paired tests, power, bankroll risk) use every holdout match. The same logistic regression 6% edge rule therefore places 139 bets on the full holdout but 77 on the closing-odds subset. A bet depends only on its own match, so the 77 are a subset of the 139, and the two ROI figures differ only because of which matches are included.

6.1 Do the models add information beyond the market?

source	metric	mean	delta_vs_market	ci_lo	ci_hi	p_value
Market (Avg, de-vigged)	log_loss	0.961	0.000	0.000	0.000	-
Market (Avg, de-vigged)	brier	0.571	0.000	0.000	0.000	-
Market (Avg, de-vigged)	rps	0.192	0.000	0.000	0.000	-
Pinnacle (de-vigged)	log_loss	0.962	0.001	-0.000	0.002	0.113
Pinnacle (de-vigged)	brier	0.572	0.001	-0.000	0.001	0.195
Pinnacle (de-vigged)	rps	0.192	0.000	-0.000	0.000	0.168
Pinnacle close (de-vigged)	log_loss	0.963	0.002	-0.002	0.007	0.293
Pinnacle close (de-vigged)	brier	0.573	0.002	-0.001	0.005	0.269
Pinnacle close (de-vigged)	rps	0.193	0.001	-0.000	0.002	0.137
Logistic regression	log_loss	0.964	0.004	-0.003	0.010	0.272
Logistic regression	brier	0.573	0.002	-0.002	0.006	0.339
Logistic regression	rps	0.193	0.001	-0.000	0.002	0.131
Market x strength-model blend	log_loss	0.961	0.000	0.000	0.000	0.000
Market x strength-model blend	brier	0.571	0.000	-0.000	0.000	0.254
Market x strength-model blend	rps	0.192	0.000	-0.000	0.000	0.625
XGBoost (raw)	log_loss	0.966	0.005	-0.001	0.012	0.104
XGBoost (raw)	brier	0.574	0.002	-0.002	0.007	0.237
XGBoost (raw)	rps	0.194	0.002	0.000	0.003	0.044
XGBoost (calibrated)	log_loss	0.970	0.010	0.003	0.017	0.006
XGBoost (calibrated)	brier	0.577	0.005	0.001	0.010	0.020
XGBoost (calibrated)	rps	0.195	0.003	0.001	0.005	0.015
Climatology (class freq.)	log_loss	1.070	0.109	0.085	0.134	0.000
Climatology (class freq.)	brier	0.647	0.075	0.058	0.093	0.000
Climatology (class freq.)	rps	0.228	0.036	0.028	0.044	0.000

Table 1: Proper scoring rules versus the market (lower is better).

source	ece	ece_noise_floor	p_vs_perfect	ece_H	ece_D	ece_A
Market (Avg, de-vigged)	0.021	0.020	0.405	0.044	0.023	0.032
Pinnacle (de-vigged)	0.026	0.020	0.126	0.043	0.031	0.034
Pinnacle close (de-vigged)	0.022	0.020	0.322	0.049	0.045	0.046
Logistic regression	0.012	0.021	0.953	0.031	0.049	0.037
Market x strength-model blend	0.021	0.020	0.412	0.044	0.023	0.032
XGBoost (raw)	0.023	0.021	0.302	0.035	0.036	0.031
XGBoost (calibrated)	0.023	0.020	0.336	0.034	0.036	0.042
Climatology (class freq.)	0.023	0.022	0.445	0.065	0.059	0.038

Table 2: Calibration error (ECE) against its sampling-noise floor.

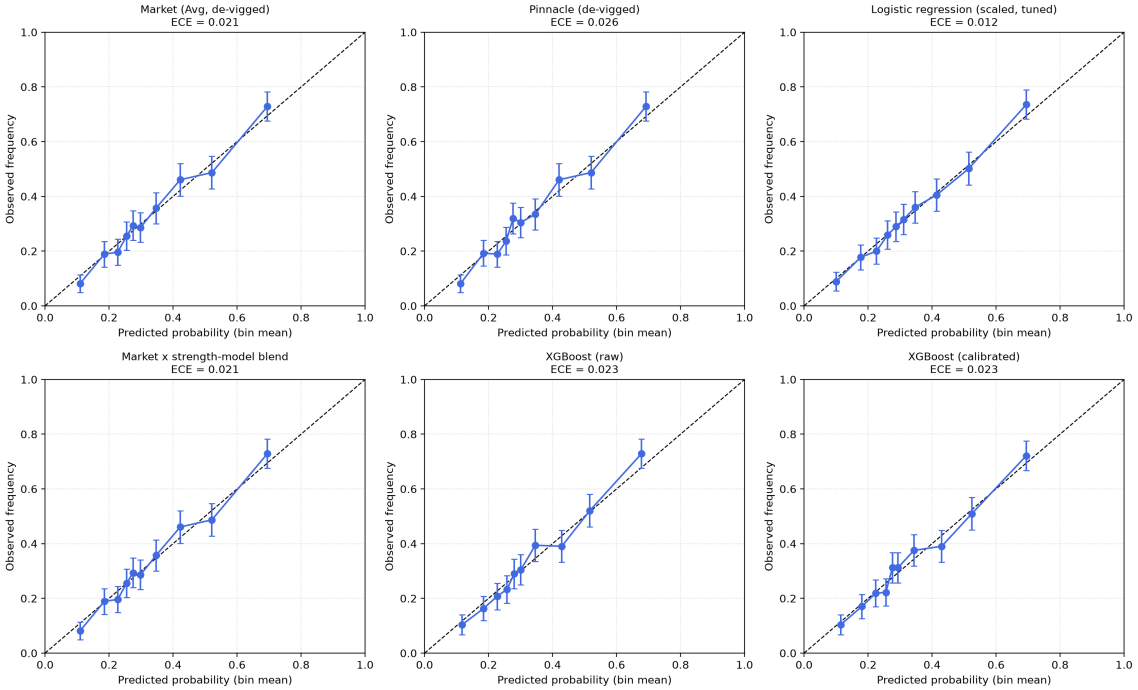


Figure 1: Reliability diagrams (equal-mass bins, pooled over H/D/A).

The market (average odds, de-vigged) achieves the best scores across all metrics (log loss 0.961, Brier 0.571, RPS 0.192). No model beats the market: the calibrated XGBoost is significantly worse (Δlog loss $= +0.010$, 95% CI $[+0.003, +0.017]$, $p = 0.006$). The logistic regression and raw XGBoost are directionally worse, with log-loss and Brier intervals that include zero; the one nominal exception is the raw XGBoost RPS ($p = 0.044$, again in the unfavourable direction). Pinnacle de-vigged prices are statistically indistinguishable from the average market. The blend's weight collapses to $w \approx 0$, confirming that Elo and form features add no information beyond market prices. Climatology is dramatically worse (Δlog loss $= +0.109$, $p < 0.001$), validating the market's informational content. All sources, including the market, are statistically compatible with perfect calibration ($p \geq 0.126$ in the ECE table), so calibration does not explain the differences in log loss.

6.2 Betting outcomes and their uncertainty

model	threshold	bets	roi	ci_lo	ci_hi	null_mean	z_vs_null	p_vs_null	ci_excludes_0
Logistic regression	0.030	860	-2.02%	-11.86%	8.14%	-7.31%	1.062	0.146	False
Market x strength-model blend	0.030	0	-	-	-	-	-	-	False
XGBoost (raw)	0.030	825	-1.04%	-12.98%	11.35%	-9.19%	1.385	0.084	False
XGBoost (calibrated)	0.030	832	-7.12%	-16.44%	2.17%	-6.98%	-0.028	0.509	False
Logistic regression	0.060	139	0.32%	-25.27%	26.62%	-7.75%	0.628	0.261	False
Market x strength-model blend	0.060	0	-	-	-	-	-	-	False
XGBoost (raw)	0.060	178	-22.85%	-45.41%	0.82%	-9.33%	-1.047	0.854	False
XGBoost (calibrated)	0.060	226	-10.58%	-27.18%	7.47%	-6.23%	-0.481	0.679	False

Table 3: ROI with bootstrap CI and efficient-market null (full holdout, 1 bet/match, edge rule).

model	league	bets	roi	ci_lo	ci_hi	p_vs_null	ci_excludes_0
Logistic regression	bundesliga	34	-16.71%	-65.74%	37.59%	0.603	False
Logistic regression	la_liga	18	-12.56%	-76.00%	57.72%	0.535	False
Logistic regression	ligue_1	19	32.37%	-32.74%	99.63%	0.119	False
Logistic regression	premier_league	16	-48.19%	-100.00%	14.75%	0.883	False
Logistic regression	serie_a	52	19.12%	-25.35%	68.00%	0.103	False
XGBoost (raw)	bundesliga	32	-27.06%	-73.60%	26.56%	0.679	False
XGBoost (raw)	la_liga	44	-60.41%	-88.41%	-26.55%	0.984	True
XGBoost (raw)	ligue_1	33	-22.88%	-69.67%	31.73%	0.672	False
XGBoost (raw)	premier_league	31	15.68%	-47.07%	85.55%	0.211	False
XGBoost (raw)	serie_a	38	-7.24%	-57.11%	49.66%	0.462	False
XGBoost (calibrated)	bundesliga	36	0.78%	-36.14%	39.14%	0.338	False
XGBoost (calibrated)	la_liga	54	-16.26%	-53.80%	26.04%	0.666	False
XGBoost (calibrated)	ligue_1	37	-31.35%	-65.05%	4.86%	0.906	False
XGBoost (calibrated)	premier_league	37	-2.70%	-46.35%	45.32%	0.413	False
XGBoost (calibrated)	serie_a	62	-4.52%	-36.60%	31.08%	0.458	False

Table 4: Per-league ROI at the 6% threshold (full holdout): note how wide the intervals are.

A	B	diff_units	ci_lo	ci_hi	p_two_sided
Logistic regression	Market x strength-model blend	0.440	-35.075	36.891	0.987
Logistic regression	XGBoost (raw)	41.120	-11.773	92.471	0.122
Logistic regression	XGBoost (calibrated)	24.340	-27.731	76.602	0.354
Logistic regression	Favorite	48.560	-42.291	137.366	0.297
Logistic regression	Random (expected)	141.253	91.736	190.694	0.000
Market x strength-model blend	XGBoost (raw)	40.680	-0.871	80.681	0.054
Market x strength-model blend	XGBoost (calibrated)	23.900	-16.561	62.630	0.233
Market x strength-model blend	Favorite	48.120	-28.262	128.962	0.226
Market x strength-model blend	Random (expected)	140.813	102.953	177.334	0.000
XGBoost (raw)	XGBoost (calibrated)	-16.780	-62.281	28.972	0.454
XGBoost (raw)	Favorite	7.440	-85.451	100.244	0.861
XGBoost (raw)	Random (expected)	100.133	49.485	150.465	0.000
XGBoost (calibrated)	Favorite	24.220	-62.061	111.326	0.566
XGBoost (calibrated)	Random (expected)	116.913	63.585	170.408	0.000
Favorite	Random (expected)	92.693	-12.275	199.487	0.088

Table 5: Paired bootstrap of total profit, A minus B in units (full holdout, 6% threshold).

model	bets	sd_per_bet	se_roi	min_detectable_roi	bets_for_2pct	bets_for_3pct	bets_for_5pct
Logistic regression	139	1.576	13.37%	33.24%	38405	17069	6145
XGBoost (raw)	178	1.560	11.70%	29.08%	37639	16729	6023
XGBoost (calibrated)	226	1.340	8.91%	22.16%	27747	12332	4440

Table 6: Power analysis (full holdout, 6% threshold): what this sample can and cannot detect.

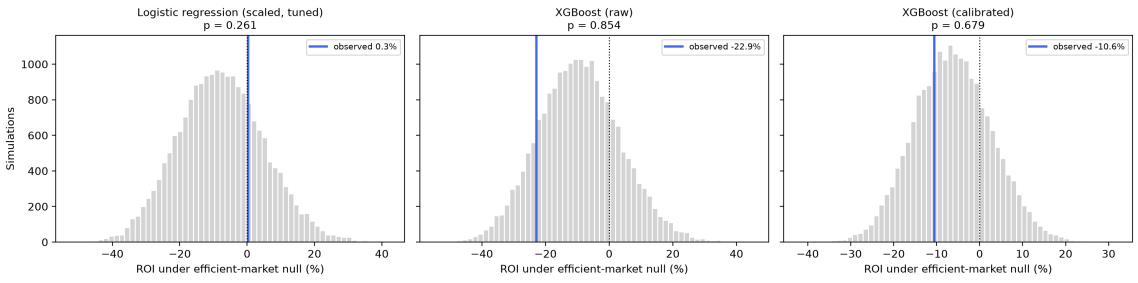


Figure 2: Observed ROI against the efficient-market null.

Across all model–threshold combinations, no strategy achieves statistical significance at the 5% level. The smallest p-value against the efficient-market null is 0.084 (XGBoost raw, 3% threshold, ROI −1.0%), which is above even the uncorrected 5% level; for the 8 strategy variants tested (two of which, the blend, place no bets) a Bonferroni threshold would be $0.05/8 \approx 0.006$. The highest point-estimate ROI in the overall table is +0.3% (logistic regression, 6% threshold), and every bootstrap interval includes zero. ROIs should be compared with the null mean (about −6% to −9%) rather than with zero: the 3% logistic regression and raw XGBoost lie 1.1 and 1.4 null standard deviations above it, which is unremarkable. Per-league ROIs rest on 16–62 bets and have intervals tens of points wide; the only interval excluding zero is raw XGBoost in La Liga, on the negative side.

The minimum detectable ROI at 80% power ranges from 22.2% (calibrated XGBoost, $n = 226$) to 33.2% (logistic regression, $n = 139$) for the 6% strategies, and is about 45% for the 77-bet selected strategy on the closing-odds subset (SE of ROI 18.0%, times $z_{0.95} + z_{0.80} = 2.49$). At the 3% threshold, with roughly 830–860 bets and $\sigma \approx 1.5$, the same formula gives about 13%. All of these are far larger than any plausible betting edge. Detecting a 2% edge would require 27,700–38,400 bets, which is roughly 30–45 times the bets placed at the 3% threshold in the holdout.

6.3 Closing-line value

strategy	odds	bets	clv	clv lo	clv hi	beat close	roi	SE ratio	skill	skill lo	skill hi	skill p
Favorite	Avg	898	-4.85%	-5.22%	-4.48%	24.16%	-	-	-	-	-	-
Random pick (expected)	Avg	898	-6.88%	-7.10%	-6.67%	-	-	-	-	-	-	-
Logistic regression (6% edge)	Avg	77	-4.12%	-5.11%	-3.08%	32.47%	7.49%	34.393	2.18%	1.29%	3.16%	0.000
XGBoost (raw) (6% edge)	Avg	65	-7.89%	-9.94%	-5.94%	23.08%	-8.60%	18.252	-0.42%	-1.86%	1.01%	0.559
XGBoost (calibrated) (6% edge)	Avg	128	-5.52%	-6.86%	-4.21%	32.03%	-20.12%	15.775	1.28%	-0.09%	2.67%	0.065
Favorite	Best	898	-2.21%	-2.70%	-1.63%	45.10%	-	-	-	-	-	-
Random pick (expected)	Best	898	-2.86%	-3.12%	-2.58%	-	-	-	-	-	-	-
Logistic regression (6% edge)	Best	77	-0.46%	-1.46%	0.55%	79.22%	-	-	2.15%	1.23%	3.12%	0.000
XGBoost (raw) (6% edge)	Best	65	-3.64%	-5.50%	-1.86%	63.08%	-	-	-0.56%	-1.89%	0.82%	0.427
XGBoost (calibrated) (6% edge)	Best	128	-2.21%	-3.47%	-0.94%	53.12%	-	-	1.01%	-0.38%	2.41%	0.153

Table 7: Closing line value against Pinnacle's closing price (898 holdout matches with Pinnacle closing odds; 6% edge rule). ROI is computed at average odds, so it is blank in the best-odds rows.

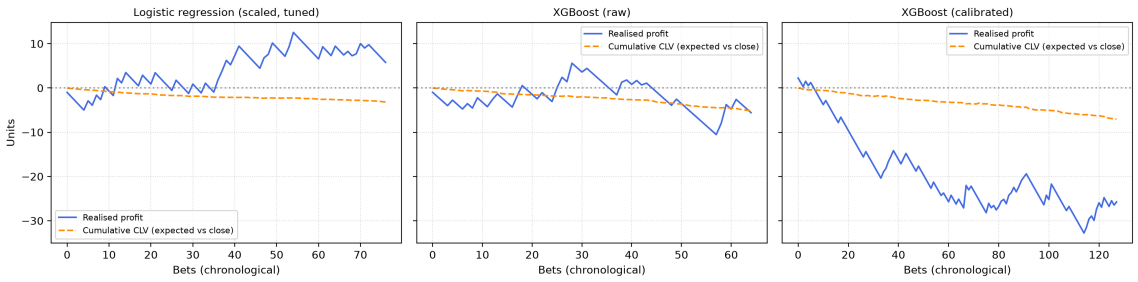


Figure 3: Realised profit versus cumulative CLV.

The pick-skill analysis isolates directional selection from the overround. For the logistic regression strategy (6% edge, average odds, 77 holdout bets), pick skill is +2.18% (95% CI $[+1.29\%, +3.16\%]$, $p < 0.001$): the model chose outcomes whose closing prices were better than the average of all three outcomes of the same match. Raw and calibrated XGBoost show no significant pick skill (−0.42% and +1.28%; $p = 0.559$ and 0.065). Two caveats apply.

First, a rule that simply backs the favourite already has a CLV of −4.85% against −6.88% for a random pick at average odds, a gap of about 2.0 points. Most of the logistic regression's

pick skill at average odds is therefore what a lean towards favourites would produce. At best available odds the favourite gap is smaller (about 0.65 points, against +2.15% for the logistic regression), so some directional skill may remain, but this was not tested separately.

Second, positive pick skill does not make CLV positive. The strategy's mean CLV is −4.12% at average odds and −0.46% (95% CI [−1.46%, +0.55%]) at best odds, and only 32.5% of wagers beat the raw closing odds at average prices. A CLV of about −4% implies an expected ROI near −4%, so the realised ROI of +7.5% is a lucky outcome of 77 bets (about 0.6 standard errors above that expectation, Figure 3) and not evidence of an edge. The variance ratio (SE of ROI / SE of CLV) is approximately 34:1 for the logistic regression (16–20:1 for the XGBoost strategies), confirming CLV's far greater precision for detecting small edges.

6.4 Strategy selected on development, evaluated on the holdout

price	Avg	Max	PS
model	Logistic regression	Logistic regression	Logistic regression
rule	edge	edge	edge
threshold	0.060	0.060	0.060
dev_bets	161	161	161
dev_clv	-2.49%	1.43%	-1.53%
hold_bets	77	77	77
hold_roi	7.49%	11.53%	9.83%
roi_lo	-25.88%	-23.92%	-25.44%
roi_hi	43.69%	49.04%	45.73%
null_mean	-7.50%	-3.76%	-4.63%
p_vs_null	0.190	0.197	0.207
hold_clv	-4.12%	-0.46%	-1.25%
clv_lo	-5.11%	-1.48%	-2.29%
clv_hi	-3.09%	0.58%	-0.21%
hold_pick_skill	2.18%	2.15%	2.21%
rank_corr_dev_vs_hold_clv	0.740	0.751	0.662

Table 8: Strategy chosen on dev (CLV), evaluated once on the holdout (898 holdout matches with Pinnacle closing odds).

The strategy selected on the 2024–25 development season (logistic regression, 6% edge threshold; 161 development bets, 77 holdout bets) behaves differently across price sources. Its mean CLV fell from −2.5% to −4.1% at average odds and from +1.4% to −0.5% at best odds, but was essentially unchanged at Pinnacle prices (−1.5% to −1.3%). The rank correlation between development and holdout CLV across the parameter grid is $\rho = 0.74$ (0.75 at best odds, 0.66 at Pinnacle prices), so the relative ordering of settings is fairly stable and the evidence for a strong winner's curse is limited. What does not carry over is the level of CLV: in the holdout it is negative at average and Pinnacle prices (the Pinnacle interval excludes zero) and indistinguishable from zero at best odds, where the development value was positive. The holdout ROI is +7.5% (95% CI [−25.9%, +43.7%]), entirely consistent with the null ($p = 0.19$). It comes from 77 bets and sits alongside a negative CLV, so it should not be read as an edge. The sample matters: on the full holdout the same rule returns +0.3% over 139 bets (Table 3), so the 62 bets on matches without closing odds returned about −8.6% ($= (139 \times 0.32\% - 77 \times 7.49\%)/62$). One rule moving from −8.6% to +7.5% between two parts of the same holdout is itself a measure of the noise, given standard errors of 13–20 points at these bet counts.

6.5 The market itself: favourite–longshot bias

bucket	n	fair p	win freq	ROI Avg	lo	hi	ROI PS	ROI Max
<1.5	2508	74.35%	75.08%	-0.60%	-2.92%	1.85%	-0.02%	2.21%
1.5-2	4401	56.16%	56.15%	-2.63%	-5.14%	-0.04%	-1.39%	0.43%
2-3	7803	39.38%	39.01%	-5.12%	-7.65%	-2.43%	-3.04%	-1.57%
3-4	12195	27.29%	27.59%	-4.66%	-7.29%	-1.92%	-2.66%	-0.03%
4-6	6768	19.66%	19.43%	-9.22%	-13.51%	-4.66%	-6.61%	-3.38%
6-10	2761	12.03%	11.34%	-17.36%	-26.23%	-8.70%	-16.19%	-9.15%
10+	938	6.12%	7.14%	-3.39%	-26.34%	20.29%	1.62%	14.11%

Table 9: Blind betting by odds bucket (favourite-longshot bias; all matches, one row per outcome).

strategy	price	n	roi	ci_lo	ci_hi
Favourite	Avg	12458	-2.57%	-4.38%	-0.86%
Favourite	PS	11601	-1.35%	-3.13%	0.40%
Favourite	Max	12458	0.62%	-1.04%	2.32%
Underdog	Avg	12458	-8.95%	-12.38%	-5.39%
Underdog	PS	11601	-6.78%	-10.40%	-3.07%
Underdog	Max	12458	-2.34%	-6.31%	1.59%
Always home	Avg	12458	-7.69%	-9.87%	-5.35%
Always home	PS	11601	-6.18%	-8.55%	-3.78%
Always home	Max	12458	-3.74%	-6.07%	-1.34%
Always draw	Avg	12458	-3.62%	-6.65%	-0.55%
Always draw	PS	11601	-1.65%	-4.73%	1.79%
Always draw	Max	12458	1.42%	-1.74%	4.67%
Always away	Avg	12458	-6.62%	-9.67%	-3.51%
Always away	PS	11601	-4.26%	-7.57%	-0.98%
Always away	Max	12458	-0.91%	-4.31%	2.41%

Table 10: Simple blind strategies over all matches.

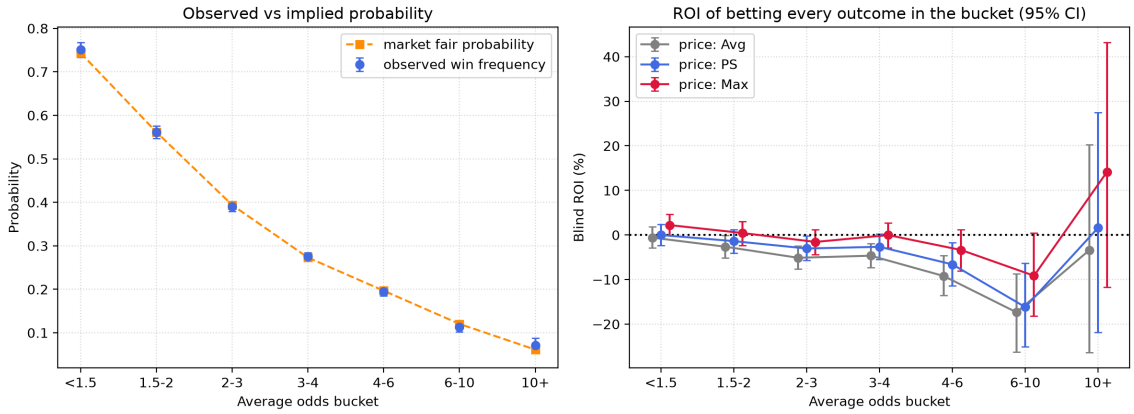


Figure 4: Observed versus implied probability and blind ROI by odds bucket.

This analysis pools 37,374 outcome-level bets, far more than the holdout. The favourite–longshot bias is statistically significant at average odds: the slope of ROI on log-odds is -0.057 (95% CI $[-0.103, -0.009]$, $p = 0.022$), meaning each unit increase in log-odds reduces ROI by about 5.7 percentage points. The loss is concentrated in the 6–10 bucket, which returns -17.4% at average odds (95% CI $[-26.2\%, -8.7\%]$), -16.2% at Pinnacle prices and -6.3% at maximum odds, while favourites ($o < 1.5$) are near break-even (-0.6% at average odds, 0.0% at Pinnacle prices, $+2.2\%$ at maximum odds). The 10+ bucket (938 outcomes) breaks the monotone pattern, with ROI between -3.4% and $+14.1\%$ depending on price source and intervals tens of points wide, so it is uninformative. The slope is smaller at Pinnacle closing prices (-0.043, $p = 0.090$) and at maximum odds (-0.011, $p = 0.684$), but both estimates lie inside the average-odds interval and the slopes were not tested against each other. We therefore do not claim the bias is absent at better prices; maximum odds, being the best across bookmakers, partly offset it by construction.

6.6 Bankroll risk

scenario	final_median	final_p5	final_p95	max_dd_median	max_dd_p5	max_dd_p95	p_final_negative
observed	-0.320	-29.971	31.832	19.620	9.810	38.082	0.506
-	-11.295	-38.974	18.607	24.640	12.000	44.782	0.732

Table 11: Flat-stake drawdown and final P&L distribution (units; logistic 6% edge strategy, full holdout).

scenario	bankroll_units	p_ruin
observed	10	0.571
observed	20	0.262
observed	50	0.004
observed	100	0.000
observed	200	0.000
-	10	0.745
-	20	0.448
-	50	0.014
-	100	0.000
-	200	0.000

Table 12: Probability of ruin by bankroll size (flat 1-unit stakes; logistic 6% edge strategy, full holdout).

scenario	scheme	final_median	final_p5	final_p95	p_loss	p_dd_over_50pct	p_below_20
observed	flat 1 unit	99.950	69.997	132.162	0.501	0.007	0.000
observed	1% of bankroll	98.234	73.121	135.227	0.537	0.000	0.000
observed	quarter Kelly (cap 3%)	104.731	62.772	181.274	0.442	0.035	0.000
-	flat 1 unit	89.170	60.605	119.181	0.725	0.020	0.000
-	1% of bankroll	88.328	66.554	118.962	0.754	0.000	0.000
-	quarter Kelly (cap 3%)	78.877	48.485	134.081	0.771	0.145	0.000

Table 13: Staking schemes from a 100-unit bankroll (logistic 6% edge strategy, full holdout).

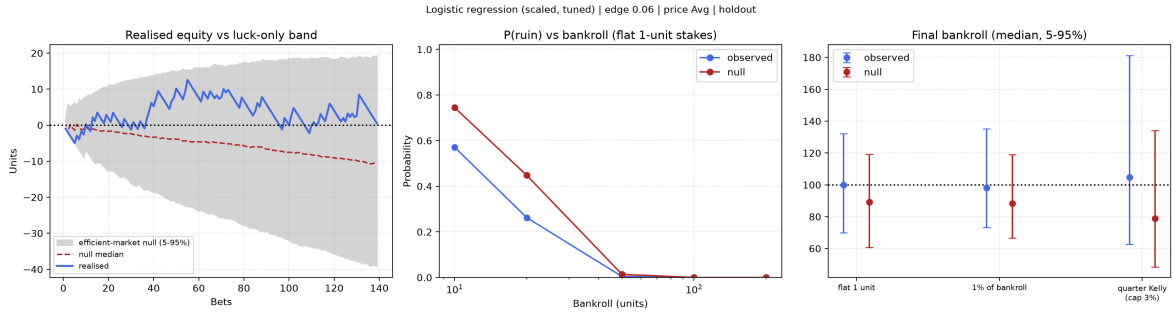


Figure 5: Equity versus the luck-only band, ruin probability by bankroll, and staking schemes.

The simulations use the 139 bets of the logistic regression 6% edge strategy on the full holdout. Under the efficient-market null, a flat 1-unit staking strategy from a 100-unit bankroll yields a median final P&L of −11.3 units (5th percentile: −39.0), with a 73.2% probability of ending negative; the "observed" scenario gives a median of −0.3 units and a 50.6% probability of ending negative. Under the null, the probability of ruin is 74.5% for a 10-unit bankroll and 44.8% for 20 units, so even "safe" flat staking carries substantial risk without an edge. Quarter-Kelly staking with model probabilities is worse: under the null it ends with a median bankroll of 78.9 units against 89.2 for flat staking, and the probability of a drawdown above 50% is 14.5% against 2.0%. A plausible mechanism is that Kelly stakes scale with the estimated edge, and the largest estimated edges are disproportionately estimation noise, so small probability errors are amplified. We did not test this explanation; the logistic regression is well calibrated on average (ECE 0.012), so any miscalibration is local rather than global.

7 Discussion and Limitations

- A single holdout cannot separate a small edge from luck; the power table states what it could detect (22–33% ROI at the 6% threshold). More data is the only remedy.
- Features exclude lineups, injuries and expected goals, and rest days ignore European and cup fixtures.
- Best-odds results assume the best quoted price can actually be obtained; average odds are a pre-closing snapshot, so CLV contains some timing noise.
- Closing odds are missing for part of the holdout, so CLV and strategy selection rest on 898 matches; whether those matches are representative (by league or period) was not checked.
- In the CLV table the ROI columns of the best-odds rows repeat the average-odds ROI, because profit is computed at average odds for both price sets; only the CLV and pick-skill columns use best odds.
- Pick skill should be read against a favourite-backing benchmark, which alone earns about 2 points at average odds.
- The favourite–longshot slopes at different price sources were not formally compared, and the 10+ odds bucket is too sparse to interpret.
- Power de-vigging is a modelling choice; Shin's method is a natural alternative.
- Rules were selected on one season; the holdout evaluates one pre-chosen strategy, but the development grid size and the number of variants tested should be borne in mind.

8 Conclusion

No model, blend, or betting strategy beats the de-vigged market with statistical support: all scoring metrics favour the market, the optimised blend weight is $w \approx 0$, and every holdout ROI is consistent with the efficient-market null. The CLV analysis reveals a small but statistically

significant pick skill (+2.2%) for the logistic regression, but much of it at average odds is what a favourite-leaning rule would give, and it does not overcome the overround: CLV is negative at average and Pinnacle prices. The power analysis shows that samples of this size (139–226 bets at the 6% threshold, 77 for the selected strategy on the closing-odds subset) cannot distinguish a realistic 2–5% edge from luck. The risk simulations show that without an edge, flat staking carries a 45–75% probability of ruin for 10–20 unit bankrolls, and quarter-Kelly staking with model probabilities increases drawdown risk relative to flat staking.

Reproducibility. `python run_all.py` regenerates every table and figure from raw data; `python -m pytest` runs the leakage and metric tests.